# MPG No. 26.5.99 $\bar{6}m'2'$ (-6m'2' setting) [ Type III, hexagonal ] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$M_z = \mathbb{M}_z^{(1)}$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)}$
$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{xxz} = 2\mathbb{M}_z^{(2)}$
$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)}$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ -M_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{yzy} = \mathbb{M}_z^{(3)}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xxyz} \\ 0 & 0 & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = \mathbb{M}_{4s}^{(1)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxyz} \\ 0 & 0 & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{T}^{(1)}_{3c}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & 0 & 0 \\ -M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{T}^{(2)}_{3c}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}^{(3)}_{3c}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 & M_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & -M_{xxyz} & 0 & 0 & -M_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & 0 & M_{xxzy} & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}^{(1)}_{4s} + 2\mathbb{T}^{(1)}_{3c} + \mathbb{T}^{(2)}_{3c}$$
$$M_{xxzy} = \mathbb{M}^{(1)}_{4s} + 2\mathbb{T}^{(1)}_{3c} - \mathbb{T}^{(2)}_{3c}$$
$$M_{yzxx} = \mathbb{M}^{(1)}_{4s} - 2\mathbb{T}^{(1)}_{3c}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}^{(3)}_{3c}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yyyz} \\ 0 & 0 & 0 \\ 0 & M_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyz} \\ 0 & 0 & 0 \\ 0 & -M_{yyyz} & 0 \\ 0 & 0 & 0 \\ -M_{yyyz} & 0 & 0 \end{bmatrix}$$

$$M_{yyyz} = -\mathbb{M}^{(1)}_{4s}$$